

Medvarsity: Transforming healthcare education with its AI-enabled cloud platform and empowering healthcare professionals

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With Google Cloud
Medvarsity has enabled
a stable, scalable
infrastructure to support
medical education
and built a data analytics

Medvarsity is Asia's largest Healthcare EdTech company with a clear focus on improving medical outcomes through better training. Our blended learning model combines online learning (synchronous & asynchronous) with bedside learning across 100+ hospital partners, to create a demonstrable improvement in the quality of care. The company is accredited by the British Accreditation Council (BAC) for Online, Distance, and Blended Learning and by CPD Standards UK for CPD credits. The brand has a wide reach, with a presence in 192 countries,

architecture design, enhance sales and improve course completions, and the user experience.

engaging an average of 3.4 million healthcare professionals monthly through its online business verticals.

Industries: Education

Location: India

Google Cloud results

- Launched a consolidated, scalable platform optimized for AI in just 10 days
- Achieved 99.99%-plus availability to ensure a high quality customer experience
- Gained the ability to leverage insights for improving a 48% course

Training healthcare professionals in 30 days

Cloud SQL (<https://cloud.google.com/sql/>),



About Brio

Brio is a technology solutions company and a Google Cloud partner that helps companies transform their business with cloud services. With offices in Hyderabad and Bangalore in India, and California in USA, Brio also delivers end-to-end custom software development services for companies.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

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completion
ratio

- Enabled the development of a large learning model that supports a guided learning experience for healthcare professionals

Compute Engine

(<https://cloud.google.com/compute/>)

and Bangalore in India and California in

Cloud Filestore (<https://cloud.google.com/filestore/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Media CDN (<https://cloud.google.com/media-cdn>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Virtual Private Cloud

(<https://cloud.google.com/vpc/>)

The education technology (edtech) market worldwide is growing rapidly, with management consulting firm McKinsey estimating (<https://www.mckinsey.com/il/podcast/educational-technology-in-the-age-of-covid>) an annual revenue of \$200 billion and growth at 15 to 20 percent per year. In healthcare, edtech is transforming the way industry professionals are educated, with Medvarsity (<https://www.medvarsity.com/>) at the forefront of that revolution.

Headquartered in India, but with a footprint in 192 countries, the business provides courses in 20 specialties online and offline for healthcare professionals such as doctors, nurses, and allied workers. These courses range from one day to two years, and most are accredited or supported by recognized healthcare industry associations,

societies, and regulators. Medvarsity also runs events and webinars, and a healthcare professional recruitment service for hospitals. Over a million healthcare professionals and other stakeholders are learning from Medvarsity's offerings.

From its foundation in 2000, Medvarsity delivered training courses using paper-based books and materials. and conducted examinations at physical locations. By 2016, however, online education had gained sufficient momentum, prompting the business to migrate to a digital model the following year. Medvarsity quickly found that its local data center could not scale to support fluctuations in demand. At the same time, the company sought to maintain high performance and uptime levels, so as to improve the user experience of its courses, webinars and other offerings. This is to encourage users to take more courses and provide positive reviews.

In addition, Medvarsity needed to ease the maintenance burden on its IT team, so they could focus on high-value tasks such as developing new products and services. After carefully reviewing cloud providers with multinational services that could support its global operations, the business opted to migrate its operations to Google Cloud (<https://cloud.google.com/>).

"My team and I are extremely tech-savvy, and when reviewing Google Cloud against another multinational cloud provider, they quickly determined Google Cloud was the natural choice for our needs," explains Gerald Jaideep, CEO of Medvarsity.

(<https://www.linkedin.com/in/geraldjaideep/>). "Its products, services, and roadmap aligned very closely with our vision for our cloud architecture and the delivery of new learning models, which will allow us to seize the opportunities presented by AI and upgraded video capabilities."

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Delivering an AI-ready cloud-based platform

Medvarsity began the first phase of the migration between 2017 and 2020, as it moved the five platforms for its service offerings to Google Cloud. It worked with Google Cloud partner Brio (<https://www.brio.co.in/>), whose team designed and implemented a solution that would drive greater scalability for Medvarsity. At the same time, Brio provided training to the Medvarsity team, providing timely support and guidance on choosing the best Google Cloud services for its business.

As part of the solution, Medvarsity hosted the platforms on Compute Engine (<https://cloud.google.com/compute>) virtual machines to

meet its requirements cost-effectively. Medvarsity also tapped on Cloud Load Balancing (<https://cloud.google.com/load-balancing>) to automate the load balancing for accommodating variations in traffic. In addition, the business migrated its database into the Cloud SQL (<https://cloud.google.com/sql>) relational database service to control costs and reduce the management load on its team. It also deployed Cloud Storage (<https://cloud.google.com/storage>) and Filestore (<https://cloud.google.com/filestore>) for highly available and durable data storage and low-latency application storage respectively.

In 2020, Medvarsity began consolidating its applications and rebuilding its platforms to become AI-ready. This included bringing the five platforms together onto one intelligent platform, which was running in Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>), to take advantage of its automated scaling capabilities. The company also moved its architecture to Virtual Private Cloud (<https://cloud.google.com/vpc>) to improve security by isolating applications and resources online, while establishing a continuous integration-continuous deployment (CI/CD) pipeline. "We are now fully automated from initial deployment onwards," says Jaideep.

The business also opted for the Flutter (<https://flutter.dev/>) open source framework for front-end application development to complement a Python-Django back end. "We've already developed a web app with Flutter, and security, performance, and

useability have all met our needs," says Jaideep.
"We're extremely pleased with the outcome."

Medvarsity released the initial iteration of its new platform, incorporating code-level API hooks to make the future deployment of AI and other new technologies seamless, to its new cloud environment in 2022. The platform is operating on an infrastructure with 99.99%-plus uptime, which minimizes the risk of disruption to services.

In late 2022, Medvarsity began preparing to build an adaptive streaming framework using Transcoder API (<https://cloud.google.com/transcoder/docs>) and Media CDN (<https://cloud.google.com/media-cdn/docs/overview>) to upgrade the video quality across its webinars and courses, and implement a range of AI use cases in its applications. With Brio providing Medvarsity with a proof of concept and a pilot, the company integrated Media CDN with Flutter, saving Medvarsity time and resources by allowing the company to experience the integration in a real-world environment. The business also reviewed the PaLM API (<https://developers.generativeai.google/>) public preview for healthcare edtech use cases that can be derived from large language models. "With PaLM API and PaLM 2 (<https://ai.google/discover/palm2/>), considerable work has been done on the healthcare segment already, which makes a huge difference to the quality of service and commitment to our users and different parts of the organization," says Jaideep.

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Medvarsity is also exploring the opportunities Vertex
AI (<https://cloud.google.com/vertex-ai>) presents to
upgrade its AI and machine learning capabilities.

Data analytics is another key focus for the business.

Medvarsity is using BigQuery

(<https://cloud.google.com/bigquery>) and AI to conduct detailed behavioral and sentiment analysis from user data captured over the last six to seven years.

"In the Asian market, up to 85 percent of healthcare professionals are connected to our platform, and we have an even larger footprint of professionals that have been associated with us at some point over the last seven years," says Jaideep. "With BigQuery and related services, we can analyze a wealth of data for trends and insights, including the education programs selected by our users, and the medical training interests they have based on their engagement with our platform."

Improving course completion ratios with detailed user insights

Armed with insights into user segments and course choice patterns, Medvarsity's student success team plans to improve a course completion ratio that already sits at a healthy 48 percent. "With data obtained through our digital services, we have a lot of call center data we can use to segmentize our customers," says Jaideep. "We will identify those users and prospects most likely to purchase, and the best offerings to promote to them."

The business also plans to upgrade the intelligence of chatbots used to support student success teams responding to user queries. "We will train a large language model across our courses and other training offerings to provide more specific answers, which will free our student success teams to deal

with more complex queries," says Jaideep. "Once a user registers as a student within our platform, we will guide them to the resources they need to learn on their own to complement their coursework. Overall, we will adopt a guided learning approach so that from the day a student starts a course, an AI-powered mechanism will identify their skillset and level, and curate a training program for them."

Launching without disruption during the pandemic

The flexibility, intuitive design and ease of use of Google Cloud enabled Medvarsity to launch its new platform without disruption during the pandemic. "Thanks to Google Cloud, we were able to launch with 120,000 users in about 10 days, with our team all working from home," says Jaideep. "This capability was particularly important as one of the highlighted courses on our platform was ventilator technician training, which was particularly important in treating people with severe cases of the virus. We were able to train over 100,000 people across India to use ventilators, which helped the country's pandemic response considerably."

With Google Cloud, Medvarsity is ideally placed to enhance the value and experience of its courses and other offerings for healthcare professionals. "We now have a very robust and secure platform, with good performance and engagement," says Jaideep. "The Google Cloud team has been extremely capable and the overall experience fantastic. It has given our business a significant boost."

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